
CMP6200/DIG6200

Individual Undergraduate Project (FYP)

Final Progress Review

Sign Language Translation System

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1 Progress to Date - Summary

The project is currently in the **hardware prototyping and data training phase**, with substantial progress made in developing the core components of the Sign Language Translation System. Since the last review, key activities initiated and completed include app development, successful integration of EMG sensors with a microcontroller, initial testing of signal acquisition using the Arduino environment and the early exploration of data interpretation techniques. In parallel, the foundational system architecture and software pipeline have been established, enabling a structured workflow from signal input to output interpretation. As this represents the first prototype, further development is planned, including an improved hardware iteration by the end of April, alongside continued work on data training and application development. The most significant finding at this stage is the identification of challenges related to signal noise and stability, which has directly influenced the project's direction by prioritising signal quality improvement and preprocessing techniques in the next phase.

2 Artefact Design and Development

2.1 Implementation and Current State

The project has successfully transitioned from the research and design phase into a functional implementation. The current prototype consists of a hardware acquisition layer and a high-fidelity mobile interface.

Completed Phases

The initial stages of the project have been successfully completed in accordance with the Gantt chart:

- **Research & Design:** Existing EMG-based systems were analyzed to identify limitations in real-time processing and signal reliability. A modular architecture was defined to separate signal acquisition from translation logic.
- **Hardware Integration:** The EMG sensor module is integrated with an Arduino Uno. Initial testing focused on power stability and grounding, which were identified as critical factors for signal integrity.
- **Software Foundation:** Basic firmware has been developed to read analog signals, while the Android application "SignLink" has been built using a modern UI framework to handle future data visualization and translation.

2.2 Technical Component Breakdown

This section explores the underlying framework of the project. Although the system is currently in a transitional state between prototyping and final optimization, the primary functional blocks are outlined below:

1) Mobile Application Interface

The frontend of SignLink is designed to be intuitive and accessible, following high-standard usability principles.

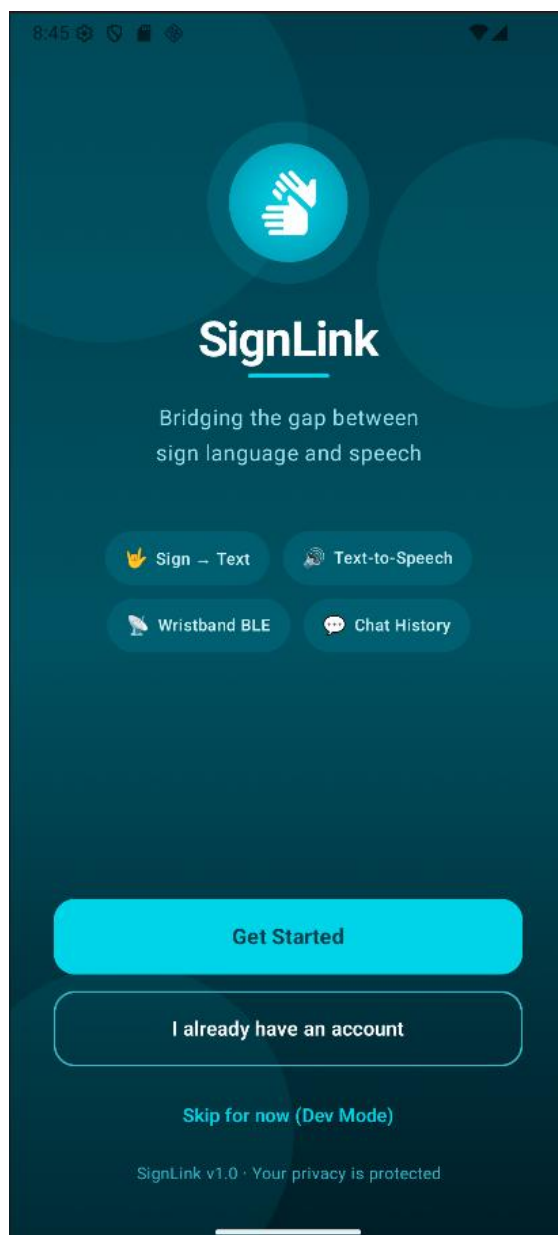


Figure 1. Start Screen

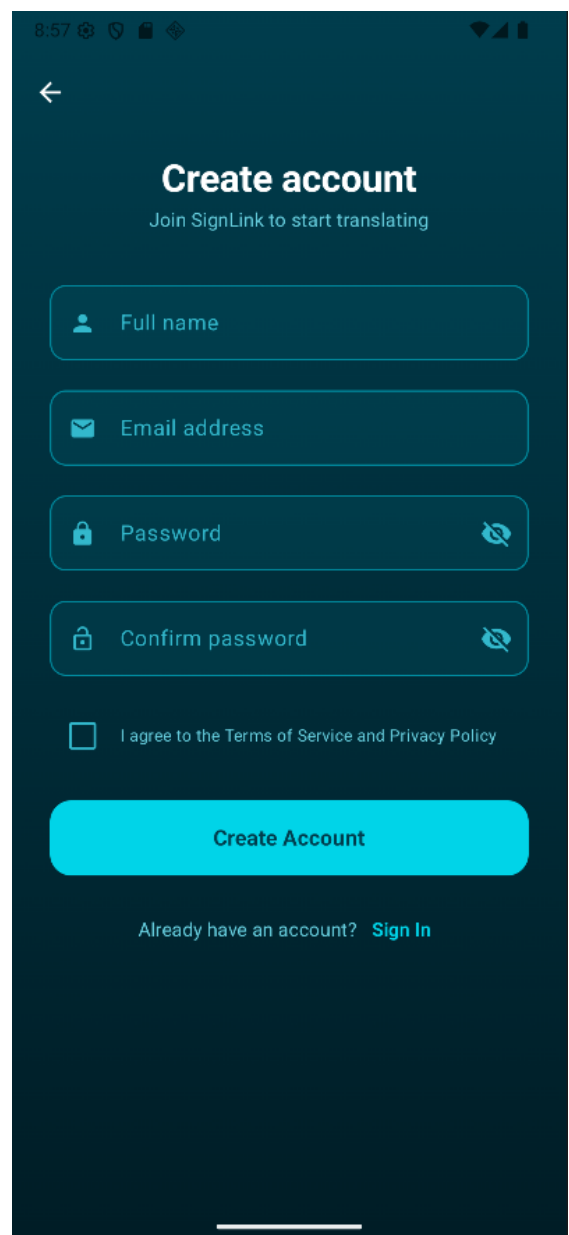


Figure 2. Sign-Up Screen

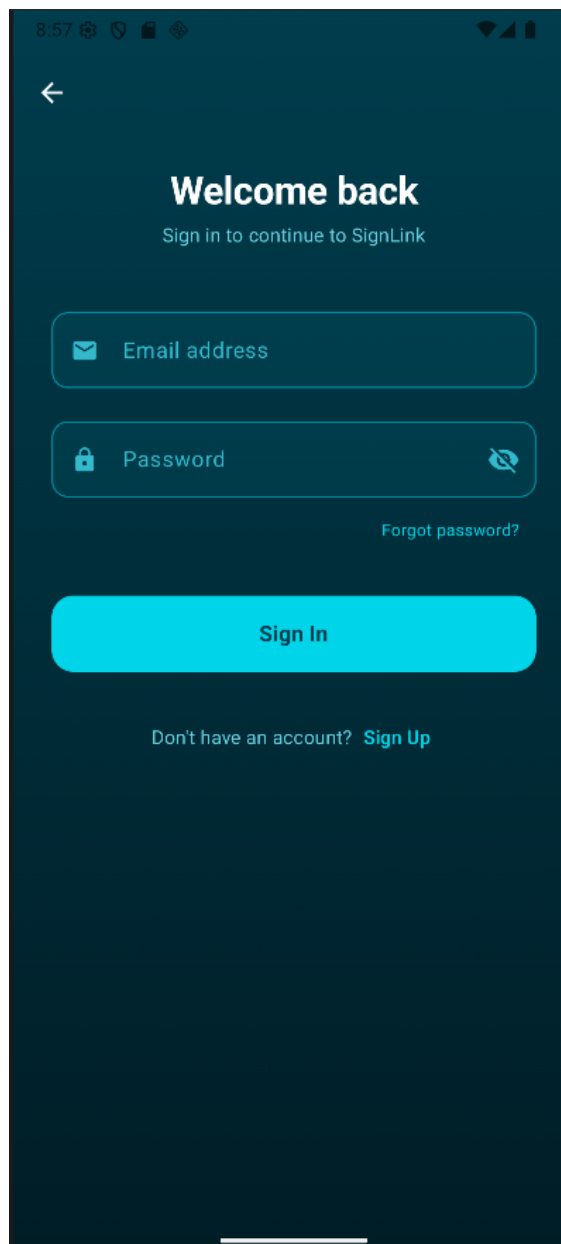


Figure 3. Sign-In Screen

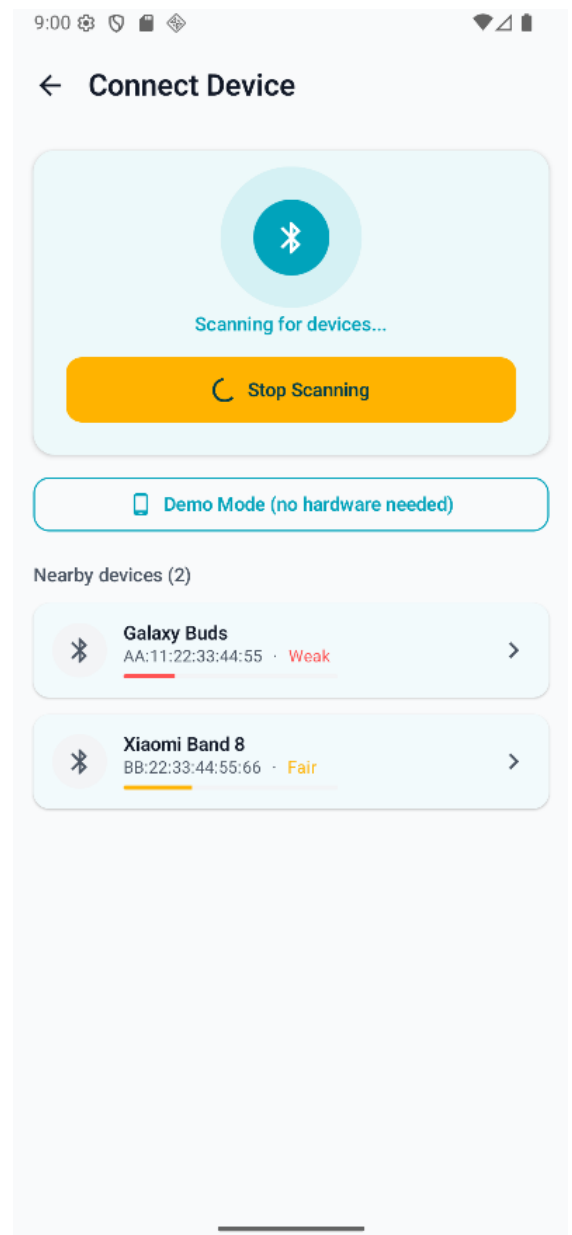
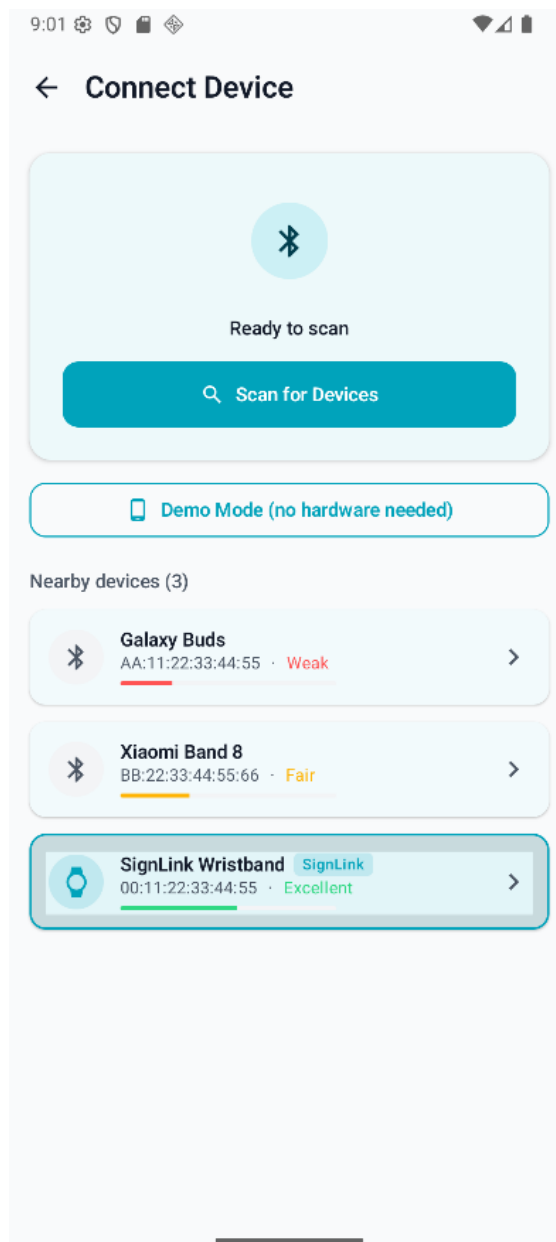
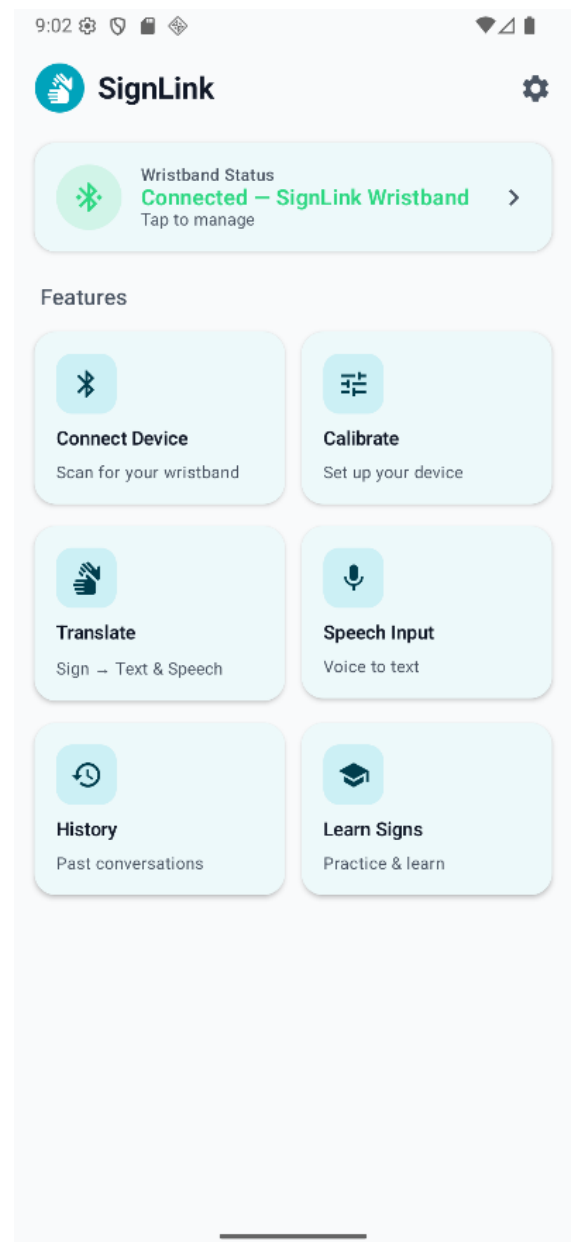


Figure 4. Device Connection Screen

**Figure 5.** Devices Spotted (Demo)**Figure 6.** Home Screen

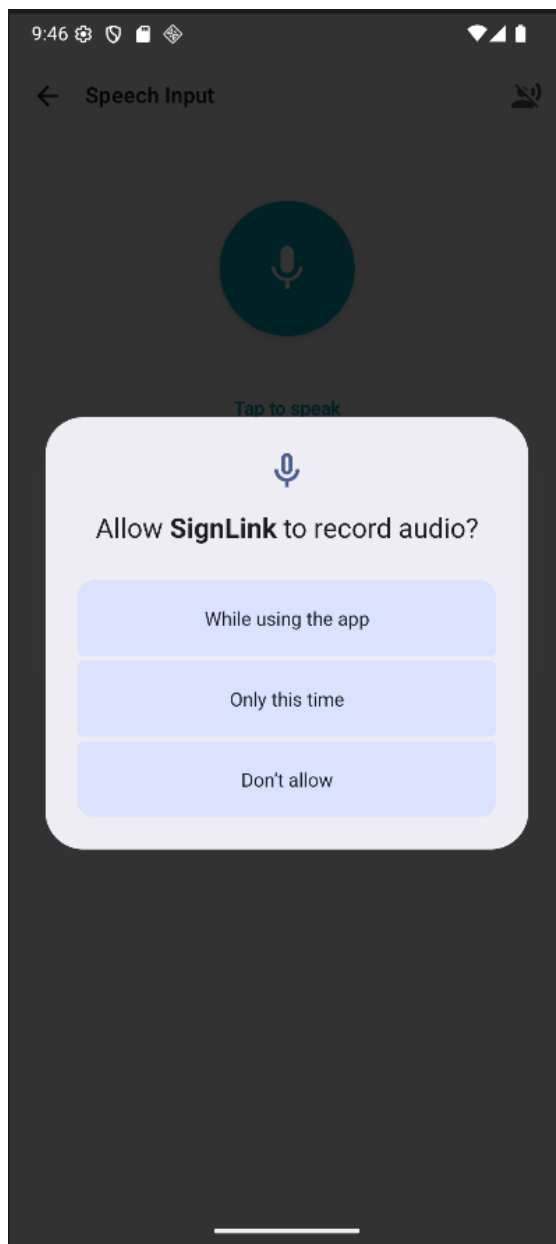


Figure 7. Voice to Text Screen (Feature)

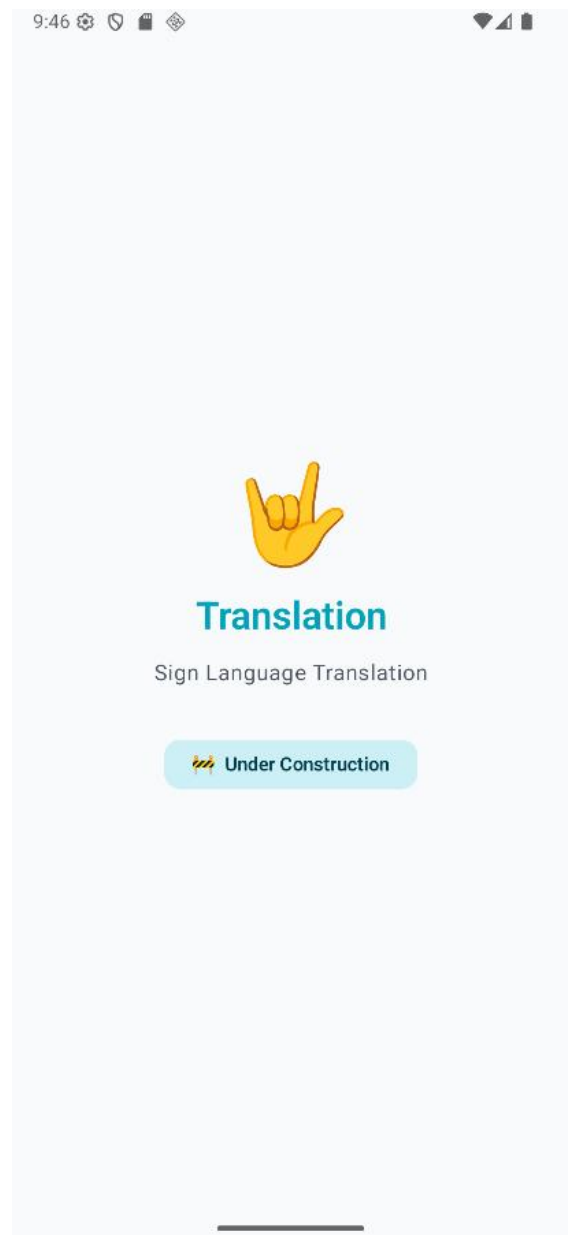
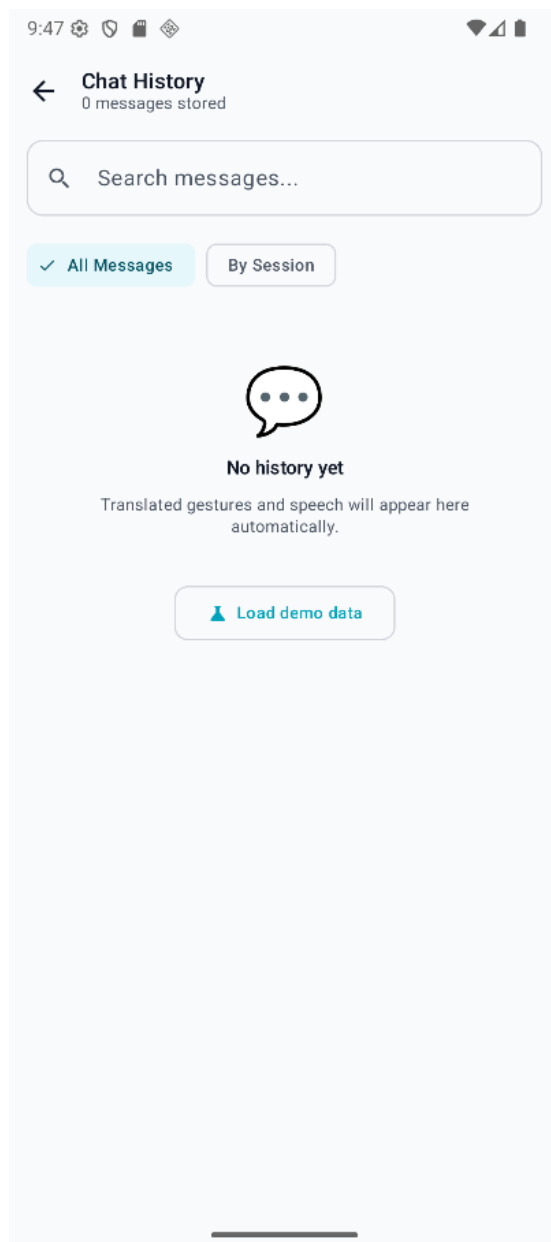
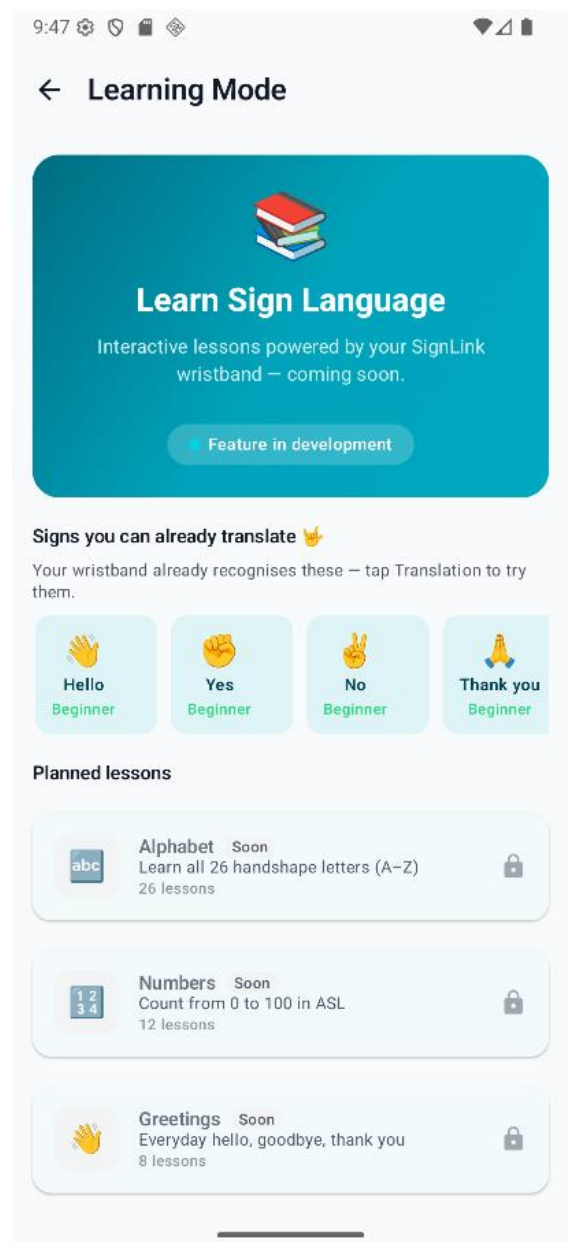


Figure 8. SignLink's Translation Screen

**Figure 9.** Chat History Screen**Figure 10.** Learn ASL Screen

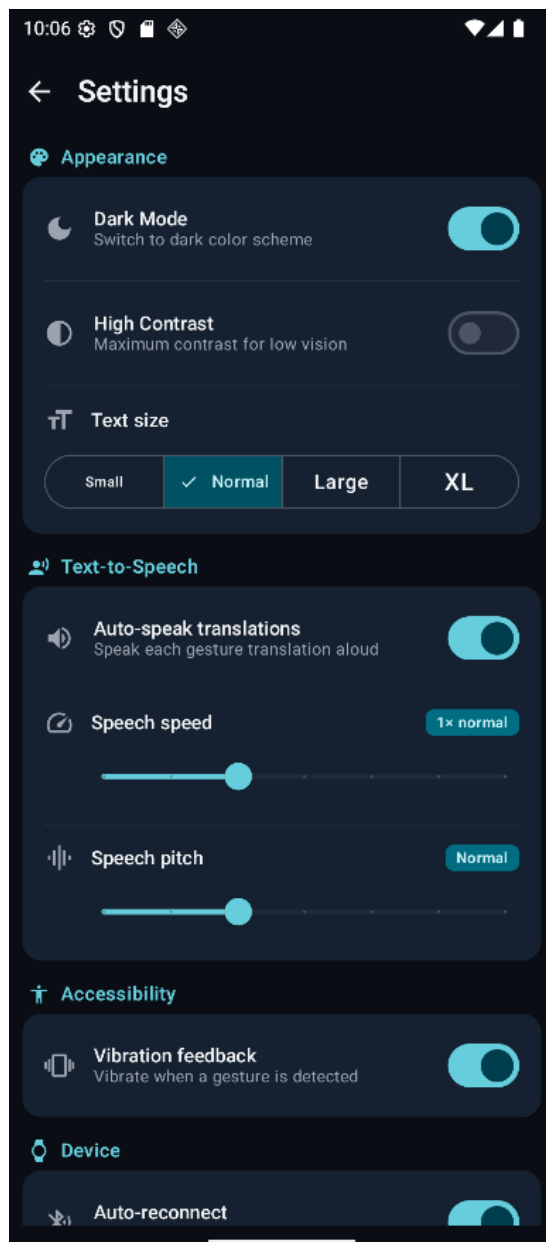


Figure 11. Settings Screen (Dark Mode has been toggled on resulting in visible color changes)

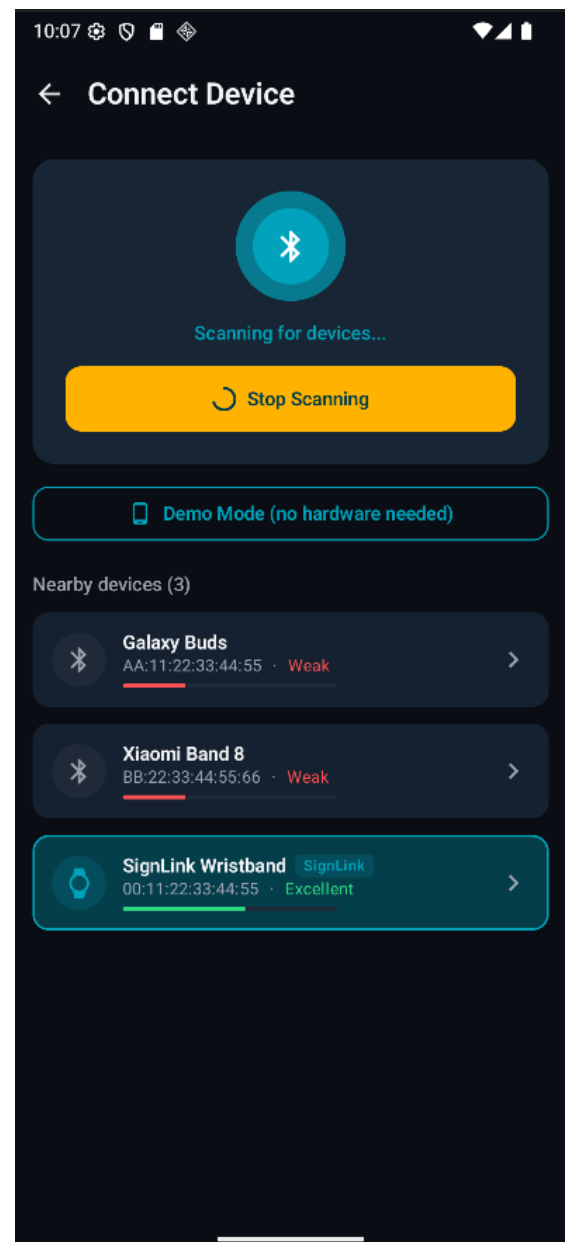


Figure 12. Device Connection Screen (Dark Mode)

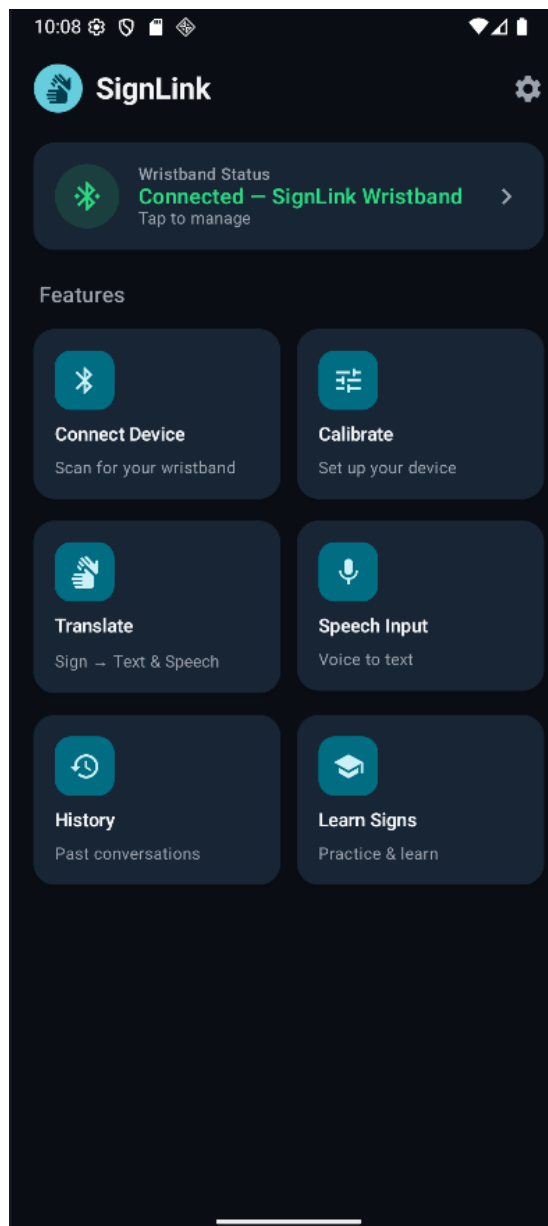


Figure 13. Home Screen (Dark Mode)

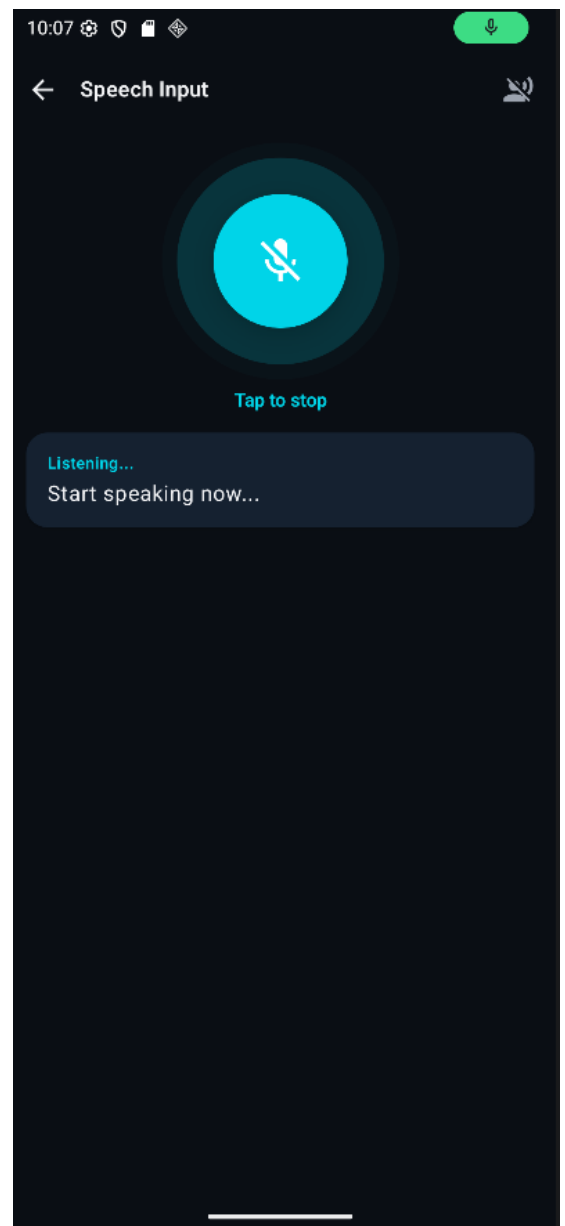


Figure 14. Voice to Text Screen (Dark Mode)

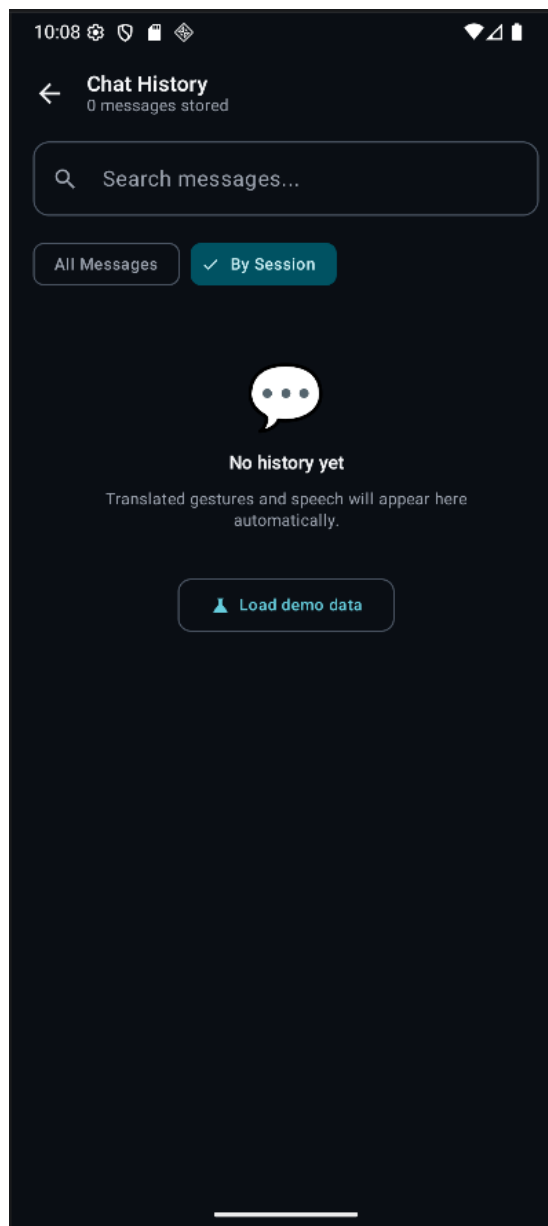


Figure 15. Chat History Screen
(Dark Mode)

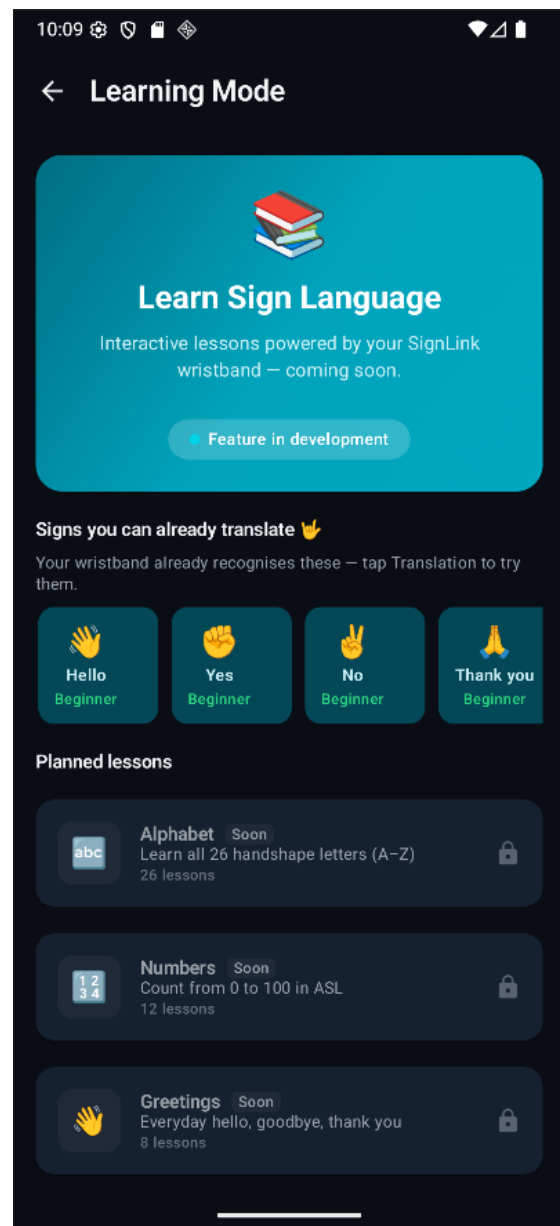


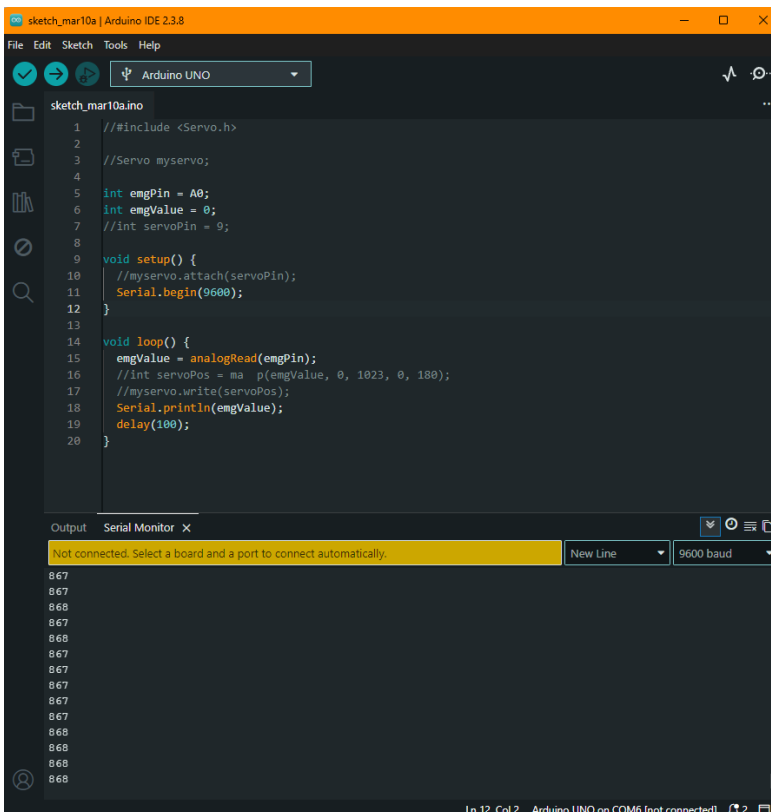
Figure 16. Learn ASL Screen
(Dark Mode)

The UX of the application was crafted after many revisions to follow Nielsen usability heuristics. This is however still a draft and therefore there will be significant changes seen in the final app after more thorough user testing among peers.

2) Connectivity and Signal Acquisition

A primary focus of the current implementation is establishing a reliable link between the wearable hardware and the mobile device.

- **Bluetooth Integration:** The app features a robust BLE scanning interface to identify and pair with the SignLink wristband.
- **Signal Capture:** Currently, EMG signals are captured and visualized via the Arduino Serial Plotter. While muscle activity is detectable, the raw data exhibits noise that requires the implementation of digital filters (e.g., Notch and Band-pass) before AI training.
- EMG signals are currently being captured and visualised using the Arduino Serial Plotter. Initial outputs confirm that the system is capable of detecting muscle activity. However, the captured signals exhibit inconsistencies and noise, indicating interference and instability in grounding.
- A key experimental observation was that disconnecting the ground wire led to significant signal fluctuations, highlighting the importance of proper grounding for accurate signal acquisition. This serves as practical validation of the system's sensitivity to electrical stability.



```

1 //include <Servo.h>
2
3 //Servo myservo;
4
5 int emgPin = A0;
6 int emgValue = 0;
7 //int servoPin = 0;
8
9 void setup() {
10   //myservo.attach(servoPin);
11   Serial.begin(9600);
12 }
13
14 void loop() {
15   emgValue = analogRead(emgPin);
16   //int servoPos = map(emgValue, 0, 1023, 0, 180);
17   //myservo.write(servoPos);
18   Serial.println(emgValue);
19   delay(100);
20 }
  
```

Output Serial Monitor X

Not connected. Select a board and a port to connect automatically.

New Line 9600 baud

Ln 12, Col 2 Arduino UNO on COM6 [not connected]

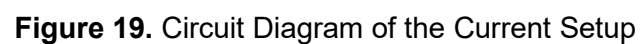
This is the code to view signals. This code does **not** specify what each high and low indicates.

As seen below, the signals are in graph format. However, they are still noisy and need a lot of filtering and cleaning work before they can be recorded and put in a dataset to train the AI.

Figure 17. Arduino Code to View Signals



The EMG sensor module has been successfully integrated with the Arduino Uno using appropriate jumper wire configurations.



During testing, issues related to power supply stability and grounding were identified and partially resolved. These findings were critical in improving the reliability of the hardware setup.

Software Components

Basic Arduino scripts have been implemented to read analog EMG signals and display them in real time. This forms the foundation for future stages involving signal filtering, feature extraction, and classification.

In addition, the Android application setup has been completed, providing a ready platform for future integration. Planned enhancements include:

- Bluetooth Low Energy (BLE) connectivity for wireless data transmission
- Integration of AI-based models for signal processing and gesture recognition

Overall, the artefact is currently in a functional prototype state, where core components are operational but require further refinement. The next stage will focus on improving signal quality, implementing processing techniques, and integrating intelligent classification methods.

2.3 Analysis and Reflection

The implementation phase has revealed a notable gap between theoretical system design and the challenges of managing biosignals in real-world scenarios. Through hands-on experimentation, several key insights emerged, altering the project's technical direction:

- **Hardware Sensitivity and Signal Integrity:** Initial tests confirmed the detectability of muscle activity, yet the raw electromyography (EMG) signals exhibited significant inconsistency due to noise and electrode sensitivity. A crucial discovery was that inadequate grounding and unstable electrical connections resulted in substantial signal fluctuations. This highlights that effective gesture recognition is not merely a software issue but is fundamentally reliant on the stability of the hardware input layer.
- **Technical Adaptation and Systematic Troubleshooting:** A critical turning point in development arose when unstable signal readings, initially thought to be caused by software bugs, were traced back to grounding issues in the hardware. This required a change in the development strategy, emphasizing hardware debugging and enhanced wiring configurations. This experience underscored the necessity of iterative testing and the importance of maintaining a controlled environment for bio-signal acquisition.
- **Validation of System Feasibility:** While the inconsistency of raw data indicates it is inadequate for direct classification, the successful capture of pertinent biological signals supports the project's fundamental premise. The potential for using EMG in sign language translation remains promising, provided a strong

preprocessing pipeline encompassing digital filtering, calibration, and noise reduction is implemented.

- **Impact on Project Direction:** The advancements made so far have effectively established the system's input layer. However, the emphasis has now shifted from rapid gesture expansion to ensuring high-quality signal acquisition. By focusing on signal preprocessing and hardware reliability, a solid foundation is being laid for the upcoming phases of feature development.

In summary, this stage of the project has bridged the gap between theory and practice, emphasizing that robust assistive technology requires a seamless integration of reliable hardware and sophisticated signal processing.

3 Updated Planning for Project Completion

To ensure the successful completion of the project within the given timeline, a series of **high-priority tasks** have been identified based on the current prototype status and remaining development phases. These tasks are prioritised according to their impact on system functionality and overall project objectives.

3.1 High-Priority Tasks

- **Refinement of Hardware Prototype**
Improve electrode placement, grounding stability, and overall circuit reliability to ensure consistent EMG signal acquisition. This is prioritised as signal quality directly affects all subsequent stages of the system.
- **Signal Processing and Preprocessing**
Implement filtering techniques (e.g., noise reduction and smoothing) and calibration methods to convert raw EMG signals into usable data. This step is essential before any classification or AI integration can be performed.
- **Data Collection and Training Dataset Development**
Collect structured EMG data for different hand gestures to build a reliable dataset. This dataset will be used to train and evaluate machine learning models for gesture recognition.
- **Development of Gesture Classification Model**
Design and implement an AI-based model capable of recognizing patterns in EMG signals and mapping them to specific gestures. This represents the core functional component of the translation system.
- **Mobile Application Integration (Android)**
Integrate the hardware system with the Android application using Bluetooth Low Energy (BLE) for wireless data transmission. Extend the application to display interpreted gestures as readable output.

- System Testing and Evaluation**
 Conduct comprehensive testing to evaluate accuracy, responsiveness, and reliability. This includes testing across different users to assess consistency and usability.
- Final Documentation and Report Writing**
 Complete the Results, Discussion, and Conclusion chapters, supported by experimental data and system evaluation.

Attached below is the Gantt chart provided for this project:

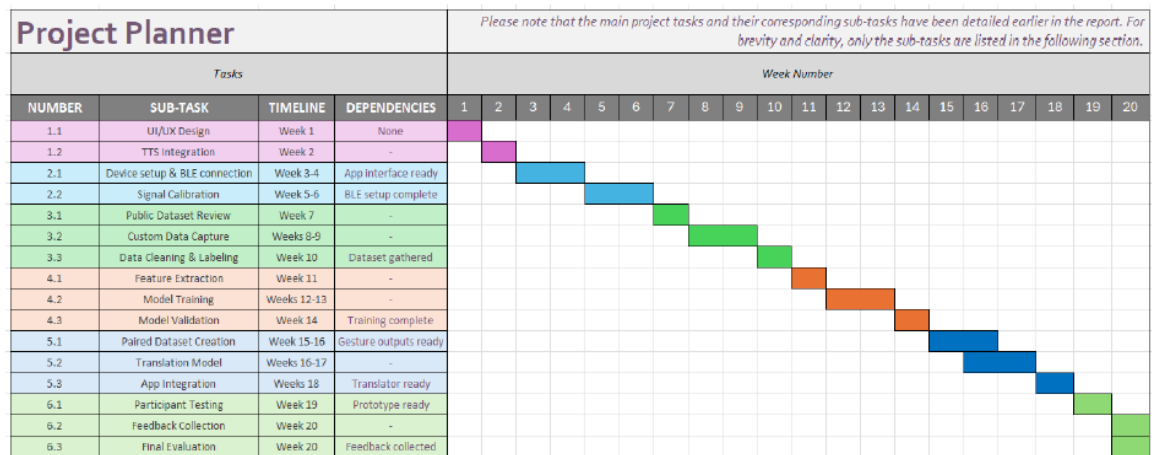


Figure 20. Initial Gantt chart

Attached further below is a rough extrapolation of the initial Gantt chart:

TASK ID	TASK TITLE	START DATE	DUE DATE	DURATION IN DAYS
STP1				
1	App Development	2-Mar-26	8-Jun-26	99
2	Write the Final Progress Review	23-Mar-26	27-Mar-26	5
	Review with advisor	1-Apr-26		
3	Setup of Hardware	25-Mar-26	27-Mar-26	3
4	Test Raw Signal Output	28-Mar-26	2-Apr-26	6
5	Fix issues and Validate Stable Readings	28-Mar-26	2-Apr-26	6
6	Define gestures / inputs and Record	3-Apr-26	7-Apr-26	5
7	Store and Label Data	3-Apr-26	7-Apr-26	5

	Review with advisor	8-Apr-26		
8	Filtering and Normalizing Signals	8-Apr-26	14-Apr-26	7
9	Train Model (Classification Logic)	12-Apr-26	20-Apr-26	9
10	Test Accuracy	17-Apr-26	24-Apr-26	8
11	Connect AI to App	22-Apr-26	29-Apr-26	8
	Review with advisor	29-Apr-26		
12	Write segment 1 and 2 of report	30-Apr-26	3-May-26	4
13	Write segment 3 and 4 of report	4-May-26	7-May-26	4
14	Setup Wearable and test signals	6-May-26	11-May-26	6
15	Fix issues and Validate Stable Readings	10-May-26	15-May-26	6
16	Define gestures / inputs and Record	15-May-26	20-May-26	6
17	Store and Label Data	20-May-26	22-May-26	3
	Review with advisor	21-May-26		
18	Retrain Model with Clean Data	23-May-26	28-May-26	6
19	Final Testing & Accuracy	27-May-26	2-Jun-26	7
20	Bug Fixes + Stability	31-May-26	4-Jun-26	5
	Review with advisor	3-Jun-26		
21	Write segment 5, 6 and 7 of report	25-May-26	7-Jun-26	14

Figure 21. Tasks and Timeline

3.2 Draft Outline of Final Report

The final report will be structured to reflect both the technical development and analytical evaluation of the project:

1. Introduction

- Background on sign language communication and accessibility challenges
- Motivation for EMG-based translation systems
- Problem statement and project objectives

2. Literature Review

- Overview of existing sign language translation technologies
- EMG-based gesture recognition systems
- Limitations of current approaches and identified research gap

3. System Architecture and Design

- Overall system architecture (hardware + software integration)
- Hardware components and selection rationale
- Data flow and system workflow design

4. Implementation

- Hardware setup and circuit configuration
- EMG signal acquisition process
- Software development (Arduino + Android application)
- BLE communication setup

5. Signal Processing and Model Development

- Signal preprocessing techniques (filtering, calibration)
- Feature extraction methods
- Machine learning model design and training process

6. Results and Evaluation

- Experimental setup and testing methodology
- Performance metrics (accuracy, latency, reliability)
- Analysis of system outputs

7. Discussion

- Interpretation of results
- Comparison with initial expectations
- Limitations of the current system

8. Conclusion and Future Work

- Summary of achievements
- Potential improvements (e.g., improved sensors, expanded gesture set, real-time optimisation)